

MORITA EQUIVALENCE OF PARTIAL MAPS AND STRONG EPIMORPHISMS

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1. INTRODUCTION

The intention of this paper is to give a rather pretty categorical proof of an equivalence of categories that seems of interest in enumerative combinatorics as per [9], representation theory as per [2], and homotopy theory as per [1].

Our results involve a category $\mathcal{A}$ equipped with a fairly general factorization system $(\mathcal{E}, \mathcal{M})$ in the sense of [7]. Pullbacks of $\mathcal{M}$ s along arbitrary morphisms should exist, leading to a category $\mathcal{P}$ of partial maps in $\mathcal{A}$. We obtain a fully faithful functor

$$\widehat{(-)} : [\mathcal{E}, \text{Mod}^R] \longrightarrow [\mathcal{P}, \text{Mod}^R]$$

with a right adjoint and monomorphic counit. If $\mathcal{A}$ is finitely $\mathcal{M}$ -well powered (that is, if every object only has finitely many $\mathcal{M}$ -subobjects), we prove that this functor is an equivalence.

Let $\mathfrak{S}$ be the groupoid of finite sets and bijective functions. Let $\text{FI}\sharp$ denote the category of finite sets and injective partial functions. Let Mod^R denote the category of left modules over the ring R . Our purpose is to understand and generalize the classification theorem for $\text{FI}\sharp$ -modules appearing as Theorem 2.24 of [2], which provides an equivalence between the category of functors $\mathfrak{S} \longrightarrow \text{Mod}^R$ and the category of functors $\text{FI}\sharp \longrightarrow \text{Mod}^R$. Their result gives a new viewpoint on representations of the symmetric groups, also a new viewpoint on Joyal species [9, 10].

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2. THE KERNEL MODULE

Let us begin with any category $\mathcal{A}$ equipped with a prefactorization system $(\mathcal{E}, \mathcal{M})$ in the sense of [7]. We regard both $\mathcal{E}$ and $\mathcal{M}$ as subcategories of $\mathcal{A}$ having the same objects as $\mathcal{A}$ but only the respective morphisms of their class. Recall that, for a morphism $e : B \rightarrow A$ in $\mathcal{E}$ and a morphism $m : X \rightarrow Y$ in $\mathcal{M}$, the following square is a pullback.

$$\begin{array}{ccc} \mathcal{A}(A, X) & \xrightarrow{\mathcal{A}(e, X)} & \mathcal{A}(B, X) \\ \mathcal{A}(A, m) \downarrow & & \downarrow \mathcal{A}(B, m) \\ \mathcal{A}(A, Y) & \xrightarrow{\mathcal{A}(e, Y)} & \mathcal{A}(B, Y) \end{array} \quad (2.1)$$

An $\mathcal{M}$ -*subobject* of an object X of $\mathcal{A}$ is an isomorphism class, in the slice category $\mathcal{A}/X$, of objects $m : U \rightarrow X$ with $m \in \mathcal{M}$.

Assume that the pullback of $\mathcal{M}$ -morphisms along arbitrary morphisms exist. A span $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$ is called an $(\mathcal{M}\text{-})$ *partial map* $f : X \rightarrow Y$ when $f_0 \in \mathcal{M}$.

Let $\mathcal{P} = \text{Par}_{\mathcal{M}}\mathcal{A}$ denote the category whose objects are all those of $\mathcal{A}$ and whose morphisms are isomorphism classes $[f]$ of partial maps. Composition is that of spans: that is, by pullback. There are functors $(-)^* : \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$ and $(-)_* : \mathcal{A} \rightarrow \mathcal{P}$ which are the identity on objects. For $m : U \rightarrow X$ in $\mathcal{M}$, define $m^* = [m, U, 1_U] : X \rightarrow U$. For any $f : X \rightarrow Y$ in $\mathcal{A}$, define $f_* = [1_X, X, f] : X \rightarrow Y$. Every partial map $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$ has a factorization

$$[f] = f_{1*} \circ f_0^* .$$

For every pullback

$$\begin{array}{ccc} P & \xrightarrow{g} & B \\ n \downarrow & & \downarrow m \\ A & \xrightarrow{f} & C \end{array} \quad (2.2)$$

in $\mathcal{A}$ with m a monomorphism, the square

$$\begin{array}{ccc} P & \xrightarrow{g_*} & B \\ n^* \uparrow & & \uparrow m^* \\ A & \xrightarrow{f_*} & C \end{array} \quad (2.3)$$

commutes in $\mathcal{P}$.

There is analogue, for partial maps in place of general spans, of the result in [15] about Mackey functors. There is an equivalence between functors $T : \mathcal{P} \rightarrow \mathcal{X}$ and pairs (T^*, T_*) of functors $T^* : \mathcal{M}^{\text{op}} \rightarrow \mathcal{X}$ and $T_* : \mathcal{A} \rightarrow \mathcal{X}$ which agree on objects and satisfy the Bénabou-Roubaud-Chevalley-Beck condition on pullbacks of the form (2.2). Indeed $T^* = T \circ (-)^*$ and $T_* = T \circ (-)_*$.

Let Set denote the category of sets and functions. We write ParSet for $\text{Par}_{\mathcal{M}}\text{Set}$ when $\mathcal{M}$ consists of the injective functions.

Notice that, if all morphisms in $\mathcal{A}$ are in $\mathcal{M}$ (such as in the category of finite sets and injective functions), then $\mathcal{E}$ consists of the invertible morphisms; it is a groupoid.

There is a “kernel” functor

$$K : \mathcal{E}^{\text{op}} \times \mathcal{P} \longrightarrow \text{ParSet} \quad (2.4)$$

defined on objects by

$$K(A, X) = \mathcal{A}(A, X) .$$

It is defined on morphisms by

$$\begin{array}{ccc} K(A, X) = \mathcal{A}(A, X) & & \\ \downarrow K(s, f) & \swarrow \mathcal{A}(A, f_0) & \\ & \mathcal{A}(A, U) & \\ & \swarrow \mathcal{A}(s, f_1) & \\ K(B, Y) = \mathcal{A}(B, Y) & & \end{array} \quad (2.5)$$

where $s : B \longrightarrow A$ is in $\mathcal{E}$ and $[f] = [f_0, U, f_1] : X \longrightarrow Y$ is a partial map in $\mathcal{A}$. To see that composition is preserved, take $t : C \longrightarrow B$ in $\mathcal{E}$ and a partial map $[g] = [g_0, V, g_1] : Y \longrightarrow Z$, and form the pullback P of f_1 and g_0 as in the diamond in the diagram:

$$\begin{array}{ccccc} & & P & & \\ & \swarrow h_0 & & \searrow h_1 & \\ & U & & V & \\ & \swarrow f_0 & & \searrow g_0 & \\ X & & Y & & Z \end{array} \quad (2.6)$$

Then we have

$$\begin{array}{ccccc} & & \mathcal{A}(A, P) & & \\ & \swarrow \mathcal{A}(A, h_0) & & \searrow \mathcal{A}(st, h_1) & \\ & \mathcal{A}(A, U) & & \mathcal{A}(B, V) & \\ & \swarrow \mathcal{A}(A, f_0) & & \searrow \mathcal{A}(t, g_1) & \\ \mathcal{A}(A, X) & & \mathcal{A}(B, Y) & & \mathcal{A}(C, Z) \end{array} \quad (2.7)$$

where the diamond is a pullback obtained as a pasting of two pullbacks, one pullback is (2.5) for $s \in \mathcal{E}$ and $p_0 \in \mathcal{M}$, and the other is the image under $\mathcal{A}(B, -)$ of the diamond pullback in (2.6). That is,

$$K(t, [g]) \circ K(s, [f]) = [\mathcal{A}(A, h_0), \mathcal{A}(A, P), \mathcal{A}(st, h_1)] = K(st, [g] \circ [f]) .$$

So K preserves composition. That K preserves identity morphisms is clear.

We write

$$\overline{K} : \mathcal{E}^{\text{op}} \longrightarrow [\mathcal{P}, \text{ParSet}]$$

for the functor corresponding to (2.4) using cartesian closedness of CAT. On objects, $\overline{K}(A) = K(A, -)$.

Note that, in ParSet, we can identify morphisms $[f] = [f_0, U, f_1] : X \longrightarrow Y$ with their canonical unique representative f of the form (f_0, U, f_1) where $U \subseteq X$ (called the *domain of definition* of f) and f_0 is the inclusion.

There is a well known equivalence of categories

$$1 + - : \text{ParSet} \simeq 1/\text{Set} \quad (2.8)$$

where $1/\text{Set}$ is the category of pointed sets and point-preserving functions. The equivalence $1 + -$ takes a set X to the set $1 + X$ which is X with a disjointly added point 0 and takes a partial function $f : X \rightarrow Y$ to the function $1 + f : 1 + X \rightarrow 1 + Y$ defined by

$$(1 + f)(x) = \begin{cases} 0 & \text{for } x \notin U \\ f(x) & \text{for } x \in U \end{cases},$$

where U is the domain of definition of f .

The equivalence (2.8) makes it clear that ParSet is complete and cocomplete. For later purposes, we remind the reader that coproducts in $1/\text{Set}$ are formed by taking the disjoint union and then identifying all the distinguished points 0. Moreover, $1/\text{Set}$ is a monoidal category whose tensor product is the smash product $(1 + S) \wedge (1 + X) = 1 + S \times X$; (this is not the categorical product in $1/\text{Set}$).

Since $[\mathcal{E}, 1/\text{Set}]$ is the free cocompletion of the free $1/\text{Set}$ -enriched category on $\mathcal{E}^{\text{op}}$ and since $[\mathcal{P}, 1/\text{Set}]$ is a cocomplete $1/\text{Set}$ -enriched category, the functor $\overline{K}$ and the equivalence (2.8) determine a colimit-preserving (pointed) functor

$$\widehat{(-)} : [\mathcal{E}, 1/\text{Set}] \rightarrow [\mathcal{P}, 1/\text{Set}]. \quad (2.9)$$

We now look in more detail at (2.9) and its right adjoint

$$\widetilde{(-)} : [\mathcal{P}, 1/\text{Set}] \rightarrow [\mathcal{E}, 1/\text{Set}]. \quad (2.10)$$

3. THE TRANSFORM ADJUNCTION

We continue with a category $\mathcal{A}$ and prefactorization system as in Section 2; however we now assume $(\mathcal{E}, \mathcal{M})$ is a factorisation system [7] for which each $m \in \mathcal{M}$ is a monomorphism. So now every morphism $f : A \rightarrow X$ in $\mathcal{A}$ factors as $f = me$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$. It follows that this factorisation is unique up to isomorphism. Moreover, the $\mathcal{M}$ -subobjects of each X is partially ordered: the order on $\mathcal{M}$ -subobjects is induced by existence of a morphism in $\mathcal{A}/X$. It also follows that $g \circ f \in \mathcal{E}$ implies $g \in \mathcal{E}$.

We abuse notation by writing $U \leq X$ for an $\mathcal{M}$ -subobject of X . If necessary, we feel free to choose a representative of the isomorphism class and denote it by $i_U : U \rightarrow X$. For each $f : X \rightarrow Y$ and $U \leq X$, define $fU \leq Y$ by $f \circ i_U = i_{fU} \circ e$ where $e \in \mathcal{E}$. For each $f : X \rightarrow Y$ and $V \leq Y$, define $f^{-1}V \leq X$ by taking $i_{f^{-1}V}$ to be the pullback of i_V along f .

Morphisms of $\mathcal{P}$ will be written as $(U, f) : X \rightarrow Y$ where $U \leq X$ and $f : U \rightarrow Y$.

We shall define a functor

$$\widehat{(-)} : [\mathcal{E}, 1/\text{Set}] \rightarrow [\mathcal{P}, 1/\text{Set}] \quad (3.11)$$

and show that it agrees with (2.9).

For $F : \mathcal{E} \rightarrow 1/\text{Set}$ and $X \in \mathcal{P}$, define

$$\widehat{F}X = \sum_{A \leq X} FA \quad (3.12)$$

where the sum is the coproduct of pointed sets. For each $(U, f) : X \longrightarrow Y$ in $\mathcal{P}$, define

$$\widehat{F}(U, f) : \sum_{A \leq X} FA \longrightarrow \sum_{B \leq Y} FB \quad (3.13)$$

by taking its composite $\widehat{F}(U, f) \circ \text{in}_A$ with the injection at $A \leq X$ to be 0 unless $A \leq U$, in which case it is the composite $\text{in}_B \circ F s$ where $f \circ i_U = i_B \circ s$ with $s \in \mathcal{E}$.

Proposition 3.1. *Under the assumptions on $\mathcal{A}$, $\mathcal{E}$ and $\mathcal{M}$, the functors (2.9) and (3.11) are isomorphic.*

Proof. Since (3.11) preserves colimits, it suffices to prove the functors agree on objects of the form $F_Z = 1 + \mathcal{E}(Z, -)$. Using factorisation, we have

$$\widehat{F}_K X = 1 + \sum_{A \leq X} \mathcal{E}(Z, A) \cong 1 + \mathcal{A}(Z, A) \cong 1 + (\overline{K}Z)X ,$$

showing that (2.9) and (3.11) agree on such objects. We leave the check on morphisms to the reader. $\square$

Corollary 3.2. *For all $F : \mathcal{E} \longrightarrow 1/\text{Set}$, there is a natural isomorphism*

$$\widehat{F}X \cong \int^{A \in \mathcal{E}} (1 + K(A, X)) \wedge FA .$$

Suppose we have a functor $T : \mathcal{P} \longrightarrow 1/\text{Set}$. Let us say $x \in TX$ is *supported* when, for all $m : U \longrightarrow X$ in $\mathcal{M}$, if $(Tm^*)x \neq 0$ then m is invertible. Define

$$\widetilde{TX} = \{x \in TX : x \text{ is supported}\} . \quad (3.14)$$

Notice that 0 is always supported. Also, it suffices to test supportedness using inclusions for m .

Lemma 3.3. *For all $s : X \longrightarrow Y$ in $\mathcal{E}$, the function $Ts_* : TX \longrightarrow TY$ restricts to give a function $\widetilde{Ts} : \widetilde{TX} \longrightarrow \widetilde{TY}$.*

Proof. We need to see that if $x \in TX$ is supported then so is $(Ts_*)x \in TY$. Take any $n : V \longrightarrow Y$ in $\mathcal{M}$ such that $(Tn^*)(Ts_*)x \neq 0$. Form the pullback

$$\begin{array}{ccc} U & \xrightarrow{q} & V \\ m \downarrow & & \downarrow n \\ X & \xrightarrow{s} & Y \end{array} \quad (3.15)$$

in $\mathcal{A}$. Using $(Tn^*)(Ts_*) = (Tq_*)(Tm^*)$, we see that $(Tm^*)x \neq 0$. Since x is supported, m is invertible. So $nq \in \mathcal{E}$ and hence $n \in \mathcal{E}$. So n is invertible. This shows that $(Ts_*)x \in \widetilde{TY}$. $\square$

Theorem 3.4. *The functor (3.11) has a right adjoint taking the functor $T : \mathcal{P} \longrightarrow 1/\text{Set}$ to $\widetilde{T}$. The unit of the adjunction is invertible; in other words, (3.11) is fully faithful. The counit of the adjunction is a monomorphism.*

Proof. We must construct a natural bijection

$$[\mathcal{P}, 1/\text{Set}](\widehat{F}, T) \cong [\mathcal{E}, 1/\text{Set}](F, \widetilde{T}) . \quad (3.16)$$

Consider a natural transformation $\theta : \widehat{F} \Rightarrow T$. Let $\theta_{X,A} : FA \rightarrow TX$ denote the restriction of θ_X along the $(A \leq X)$ -component of the injection. Observe that $\theta_{A,A} : FA \rightarrow TA$ lands in $\widetilde{TA}$. For, suppose we have a $U \leq A$ with $(Ti_U^*)\theta_{A,A}a \neq 0$ invertible for some $a \in FA$. Then, by naturality, $\theta_{A,U}(\widehat{Fi}_U^*)a \neq 0$. Of course, then $(\widehat{Fi}_U^*)a \neq 0$, and by definition of $\widehat{F}$ on morphisms, $A \leq U$. So $U = A$, as required.

Now we can define $\phi_A : FA \rightarrow \widetilde{TA}$ by $\phi_A a = \theta_{A,A}a$. We shall show that these morphisms are the components of a natural transformation $\phi : F \Rightarrow \widetilde{T}$. Take any $s : A \rightarrow B$ in $\mathcal{E}$. Then

$$(\widetilde{T}s)\phi_A a = (Ts_*)\theta_{A,A}a = \theta_B(\widehat{F}s_*)\text{in}_A a = \theta_{B,B}(Fs)a = \phi_B(Fs)a$$

using the naturality of θ . So we have a natural function $\theta \mapsto \phi$ for (3.16).

For the inverse bijection, take a natural transformation $\phi : F \rightarrow \widetilde{T}$ and define $\theta_X : \widehat{F}X \rightarrow TX$ by

$$\theta_{X,A}a = (Ti_{A*})\phi_A a .$$

We prove naturality of θ in two steps, first with respect to morphisms i_U^* for $U \leq X$ and then with respect to f_* for $f : X \rightarrow Y$ in $\mathcal{A}$. For the first, form the pullback span $(u, W, v) : A \rightarrow V$ of the pair $i_A : A \rightarrow X$ and $i_U : U \rightarrow X$. Then

$$(Ti_U^*)\theta_{X,A}a = (Ti_U^*)(Ti_{A*})\phi_A a = (Tv_*)(Tu^*)\phi_A a = T(v_*u^*)\phi_A a$$

which is 0 unless $(Tu^*)\phi_A a \neq 0$. However, $\phi_A a$ is supported, so u is invertible. On the other hand, $\theta_U(\widehat{Fi}_U^*)\text{in}_A a = 0$ unless $A \leq V$ with representative $j : A \rightarrow V$, say, in which case

$$\theta_U(\widehat{Fi}_U^*)\text{in}_A a = \theta_{U,A}a = (Tj_*)\phi_A a = T(v_*u^*)\phi_A a .$$

Now we turn to naturality of ϕ with respect to f_* . This requires factoring $fi_A = i_Bs$ with $s \in \mathcal{E}$. Then

$$\begin{aligned} \theta_Y(\widehat{F}f_*)a &= \theta_Y(Fs)a = (Ti_{B*})\phi_B(Fs)a = (Ti_{B*})(\widetilde{T}s)\phi_A a \\ &= T(i_{B*}s_*)\phi_A a = T(f_*i_{A*})\phi_A a = (Tf_*)\theta_X a . \end{aligned}$$

It is easy to see that $\theta \mapsto \phi$ and $\phi \mapsto \theta$ are mutually inverse.

The component of the unit $\eta_F : F \Rightarrow \widehat{F}$ at A is just the injection $\text{in}_A : FA \rightarrow \widehat{F}A$ seen as landing in the supported elements; so it is injective. Take any supported $\text{in}_U x \in \widehat{F}A$. So $x \in FU$. If $x = 0$ then it is in the image of $\eta_F A$. If $x \neq 0$ then $(\widehat{Fi}_U^*)x = x \neq 0$, and, by supportedness, $U = A$; so $\text{in}_U x = \text{in}_A x$ is in the image of $\eta_F A$. So η_F is invertible.

The component of the counit $\varepsilon_T : \widehat{T} \rightarrow T$ at X takes $a \in \widehat{T}A$ to $(Ti_{A*})a \in TX$ for $A \leq X$. If $(Ti_{A*})a = 0$ then $(\widetilde{T}i_A^*)(Ti_{A*})a = 0$; but $i_A^*i_{A*} = 1_A$, so $a = 0$. If $(Ti_{A*})a = (Ti_{B*})b \neq 0$ with $a \in \widehat{T}A$, $b \in \widehat{T}B$, then $T(i_B^*i_{A*})a = T(i_B^*i_{B*})b = b$. Form the pullback (u, P, v) of i_A and i_B . Then we have $T(u_*v^*)a = T(i_B^*i_{A*})a = b \neq 0$. So $T(v^*)a \neq 0$. Since a is supported, v is invertible. It follows that $A \leq B$. Symmetrically, $B \leq A$. So $A = B$ and $a = b$. This proves injectivity of $\varepsilon_T X$. So ε_T is a monomorphism. $\square$

Remark 3.5. The essential image of (3.11) consists of those T with the property that each $x \in TX$ is of the form $x = T(i_{U*})u$ for some $U \leq X$ and supported $u \in TU$.

Corollary 3.6. *For all $T : \mathcal{P} \rightarrow 1/\text{Set}$, there is a natural isomorphism*

$$\tilde{T}A \cong \int_{X \in \mathcal{P}} [K(A, X), TX] .$$

There is no problem replacing $1/\text{Set}$ by the category Mod^R . The same formula (3.12) provides a functor

$$\widehat{(-)} : [\mathcal{E}, \text{Mod}^R] \rightarrow [\mathcal{P}, \text{Mod}^R] . \quad (3.17)$$

Similarly, the formula (3.14) gives the value at T of a right adjoint

$$\widetilde{(-)} : [\mathcal{P}, \text{Mod}^R] \rightarrow [\mathcal{E}, \text{Mod}^R] . \quad (3.18)$$

to (3.17). Notice that $\tilde{T}X$ is a submodule of TX : supported elements are closed under addition and scalar multiple (if $(Tm^*)(x + y) = (Tm^*)x + (Tm^*)y \neq 0$ then either $(Tm^*)x \neq 0$ or $(Tm^*)y \neq 0$; and, if $(Tm^*)(rx) = r(Tm^*)x \neq 0$ then $(Tm^*)x \neq 0$).

The proof of Theorem 3.4 also proves:

Theorem 3.7. *The functor (3.17) has the right adjoint (3.18). The unit of the adjunction is invertible; in other words, (3.17) is fully faithful. The counit of the adjunction is a monomorphism.*

4. TWO COMONADS

We continue with $\mathcal{A}$, $\mathcal{E}$, $\mathcal{M}$ and $\mathcal{P}$ as before.

Let $J : \mathcal{E} \rightarrow \mathcal{A}$ be the inclusion functor and let $I = (-)_* : \mathcal{A} \rightarrow \mathcal{P}$. Both functors are the identity on objects.

Let $\mathcal{X}$ be any category admitting coproducts and products indexed by sets of $\mathcal{M}$ -subobjects of any given object of $\mathcal{A}$.

Lemma 4.1. *Each functor $F : \mathcal{E} \rightarrow \mathcal{X}$ has a pointwise left Kan extension*

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{J} & \mathcal{A} \\ & \searrow F & \swarrow \text{Lan}_J F \\ & \mathcal{X} & \end{array} \quad \begin{array}{c} \xrightarrow{\kappa} \\ \xleftarrow{\quad} \end{array} \quad (4.19)$$

along $J : \mathcal{E} \rightarrow \mathcal{A}$ defined on objects by:

$$(\text{Lan}_J F)X = \sum_{U \leq X} FU .$$

For morphisms $f : X \rightarrow Y$ in $\mathcal{A}$, the following square commutes.

$$\begin{array}{ccc} FU & \xrightarrow{\text{in}_U} & (\text{Lan}_J F)X \\ \downarrow Fe & & \downarrow (\text{Lan}_J F)f \\ FfU & \xrightarrow{\text{in}_{fU}} & (\text{Lan}_J F)Y \end{array} \quad (4.20)$$

Proof. The inclusion $U \leq X \mapsto (U, i_U : JU \longrightarrow X)$ of the discrete category on the set $\{U : U \leq X\}$ into the comma category J/X has a left adjoint, taking $(V, f : JV \longrightarrow X)$ to $fV \leq X$, and so is final. It follows that the colimit of $J/X \xrightarrow{\text{dom}} \mathcal{E} \xrightarrow{F} \mathcal{X}$ can be calculated by restricting along the inclusion and so is the coproduct displayed. $\square$

Lemma 4.2. *Each functor $T : \mathcal{A} \longrightarrow \mathcal{X}$ has a pointwise right Kan extension*

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{I} & \mathcal{P} \\ & \swarrow T & \nwarrow \text{Ran}_I T \\ & \mathcal{X} & \end{array} \quad \begin{array}{c} \xleftarrow{\rho} \\ \end{array} \quad (4.21)$$

along $I : \mathcal{A} \longrightarrow \mathcal{P}$ defined on objects by:

$$(\text{Ran}_I T)X = \prod_{U \leq X} TU .$$

For morphisms $(W, h) : X \longrightarrow Y$ in $\mathcal{P}$, the square

$$\begin{array}{ccc} (\text{Ran}_I T)X & \xrightarrow{\text{pr}_{h^{-1}V}} & Th^{-1}V \\ (\text{Ran}_I T)(W, h) \downarrow & & \downarrow T\bar{h} \\ (\text{Ran}_I T)Y & \xrightarrow{\text{pr}_V} & TV \end{array} \quad (4.22)$$

commutes, where $\bar{h}$ is the pullback of h along i_V .

Proof. The functor $U \leq X \mapsto (i_U^* : X \longrightarrow IU, U)$ from the discrete category on the set $\{U : U \leq X\}$ into the comma category X/I has a right adjoint, taking $((W, h) : X \longrightarrow IA, A)$ to $W \leq X$, and so is initial. It follows that the limit of $X/I \xrightarrow{\text{cod}} \mathcal{A} \xrightarrow{T} \mathcal{X}$ can be calculated as the product displayed. $\square$

This gives the two adjunctions

$$[\mathcal{E}, \mathcal{X}] \xrightleftharpoons[\begin{smallmatrix} \perp \\ [J, 1] \end{smallmatrix}]{\begin{smallmatrix} \text{Lan}_J \\ \end{smallmatrix}} [\mathcal{A}, \mathcal{X}] \xrightleftharpoons[\begin{smallmatrix} \top \\ [I, 1] \end{smallmatrix}]{\begin{smallmatrix} \text{Ran}_I \\ \end{smallmatrix}} [\mathcal{P}, \mathcal{X}] . \quad (4.23)$$

Each adjunction generates a comonad on $[\mathcal{A}, \mathcal{X}]$:

$$G = \text{Lan}_J \circ [J, 1] \quad \text{and} \quad H = [I, 1] \circ \text{Ran}_I . \quad (4.24)$$

To be more explicit, the functors GF and HF are defined on objects by

$$(GF)A = \sum_{U \leq A} FU \quad \text{and} \quad (HF)A = \prod_{U \leq A} FU .$$

On morphisms $f : A \longrightarrow B$ in $\mathcal{A}$, they are defined by commutativity in the squares

$$\begin{array}{ccc} FU & \xrightarrow{Fe} & FfU \\ \text{in}_U \downarrow & & \downarrow \text{in}_{fU} \\ (GF)A & \xrightarrow{(GF)f} & (GF)B \end{array} \quad \begin{array}{ccc} (HF)A & \xrightarrow{(HF)f} & (HF)B \\ \text{pr}_{f^{-1}V} \downarrow & & \downarrow \text{pr}_V \\ Ff^{-1}V & \xrightarrow{F\bar{f}} & FV \end{array} . \quad (4.25)$$

The counits $\varepsilon_F : GF \rightarrow F$ and $\varepsilon_F : HF \rightarrow F$ have components respectively defined by $\varepsilon_F A \circ \text{in}_U = F\text{in}_U$ and $\varepsilon_F A = \text{pr}_A$. The comultiplications $\delta_F : GF \rightarrow G^2F$ and $\delta_F : HF \rightarrow H^2F$ have components respectively defined as follows.

$$\begin{array}{ccc}
 FU & \xrightarrow{\text{in}_U} & \sum_{U \leq A} FU \\
 \searrow \text{in}_{U \leq U} & & \swarrow \delta_F A \\
 & \sum_{V \leq U \leq A} FV &
 \end{array}
 \qquad
 \begin{array}{ccc}
 \prod_{U \leq A} FU & \xrightarrow{\text{pr}_V} & FV \\
 \searrow \delta_F A & & \swarrow \text{pr}_{V \leq U} \\
 & \prod_{V \leq U \leq A} FV &
 \end{array}$$

We shall construct a comonad morphism $\Theta : G \Rightarrow H$ when $\mathcal{X}$ is a pointed category.

Take $F : \mathcal{A} \rightarrow \mathcal{X}$ and $A \in \mathcal{A}$. We define

$$\Theta_F A : \sum_{U \leq A} FU \rightarrow \prod_{V \leq A} FV \quad (4.26)$$

by taking the composite

$$FU \xrightarrow{\text{in}_U} \sum_{U \leq A} FU \xrightarrow{\Theta_F A} \prod_{V \leq A} FV \xrightarrow{\text{pr}_V} FV$$

to be zero unless $U \leq V$, in which case the composite is $F(U \leq V) : FU \rightarrow FV$.

Proposition 4.3. *For the two comonads G and H of (4.24) on $[\mathcal{A}, \mathcal{X}]$, a comonad morphism $\Theta : G \Rightarrow H$ is defined by the components (4.26).*

Proof. Naturality in A is proved by contemplating the following diagram.

$$\begin{array}{ccccccc}
 FU & \xrightarrow{\text{in}_U} & \sum_{U \leq A} FU & \xrightarrow{\Theta_F A} & \prod_{U_1 \leq A} FU_1 & \xrightarrow{\text{pr}_{f^{-1}V}} & Ff^{-1}V \\
 \downarrow Fe & & \downarrow (GF)f & & \downarrow (HF)f & & \downarrow F\bar{f} \\
 FfU & \xrightarrow{\text{in}_{fU}} & \sum_{V_1 \leq B} FV_1 & \xrightarrow{\Theta_F B} & \prod_{V \leq B} FV & \xrightarrow{\text{pr}_V} & FV
 \end{array}$$

The top composite is zero unless $U \leq f^{-1}V$ (say with inclusion i). The bottom composite is zero unless $fU \leq V$ (say with inclusion j). The conditions are the same. When this condition holds, the diagram commutes since $\bar{f} \circ i = j \circ e$.

Naturality in F is obvious.

Preservation of the counits is proved by the calculation

$$\varepsilon_F A \circ \Theta_F A \circ \text{in}_U = \text{pr}_A \circ \Theta_F A \circ \text{in}_U = F\text{in}_U = \varepsilon_F A \circ \text{in}_U.$$

Preservation of the comultiplications is proved by observing that

$$\text{pr}_{V \leq U} \circ \delta_F A \circ \Theta_F A \circ \text{in}_W = \text{pr}_V \circ \Theta_F A \circ \text{in}_W$$

while

$$\text{pr}_{V \leq U} \circ \Theta_F^2 A \delta_F A \circ \text{in}_W = \text{pr}_{V \leq U} \circ \Theta_F^2 A \circ \text{in}_{W \leq W}.$$

Both of these right-hand sides are zero unless $W \leq V$, in which case they are both equal to $F(W \leq V)$. $\square$

Proposition 4.4. *The functor $[I, 1] : [\mathcal{P}, \mathcal{X}] \rightarrow [\mathcal{A}, \mathcal{X}]$ is comonadic.*

Proof. The functor $[I, 1]$ is conservative since I is bijective on objects. It also preserves all limits, including all equalizers. The result follows (for example) by the Beck comonadicity theorem [16]. $\square$

It follows that $\Theta : G \Longrightarrow H$ induces a functor $\bar{\Theta} : [\mathcal{A}, \mathcal{X}]^G \longrightarrow [\mathcal{P}, \mathcal{X}]$ over $[\mathcal{A}, \mathcal{X}]$, where $[\mathcal{A}, \mathcal{X}]^G$ is the category of Eilenberg-Moore G -coalgebras. The composite of $\bar{\Theta}$ with the comparison functor $[\mathcal{E}, \mathcal{X}] \longrightarrow [\mathcal{A}, \mathcal{X}]^G$ is isomorphic to

$$\widehat{(-)} : [\mathcal{E}, \mathcal{X}] \longrightarrow [\mathcal{P}, \mathcal{X}] \quad (4.27)$$

as defined in (3.12).

5. THE MORITA EQUIVALENCE

With a finite well-poweredness assumption on the factorization system of $\mathcal{A}$, we shall show that the adjunction of Theorem 3.7 is an equivalence. In other words, we shall show that the free additive categories on $\mathcal{E}$ and $\mathcal{P}$ are Morita equivalent (= additively Cauchy equivalent).

Our goal in this section is to prove our main theorem.

Theorem 5.1. *Let $(\mathcal{E}, \mathcal{M})$ be a factorisation system on a category $\mathcal{A}$. Assume that $\mathcal{M}$ is contained in the class of monomorphisms and that all pairs $(m : U \rightarrow X, f : A \rightarrow X)$ of morphisms with $m \in \mathcal{M}$ have a pullback in $\mathcal{A}$. Assume that the $\mathcal{M}$ -subobjects of each object form a finite set. Regard $\mathcal{E}$ as a subcategory of $\mathcal{A}$ with the same objects and put $\mathcal{P} = \text{Par}_{\mathcal{M}}\mathcal{A}$. Let $\mathcal{X}$ be any finitely complete additive category. Then the functor (4.27) is an equivalence of categories*

$$[\mathcal{E}, \mathcal{X}] \simeq [\mathcal{P}, \mathcal{X}] .$$

We first prove that the comonads G and H of (4.24) are isomorphic.

Proposition 5.2. *Under the assumptions of Theorem 5.1, the comonad morphism $\Theta : G \Longrightarrow H$, with components (4.26), is invertible.*

Proof. Since $\mathcal{X}$ is additive, every finite product is a coproduct. The ordered set of $\mathcal{M}$ -subobjects of $A \in \mathcal{A}$ is assumed finite and so has a linear refinement. So we can list all the subobjects as $0 = U_0, U_1, \dots, U_n = A$ such that $0 \leq i \leq j \leq n$ implies $U_i \leq U_j$. Then the morphisms (4.26) are represented by an upper-triangular matrix with identity morphisms in the main diagonal. Since $\mathcal{X}$ is additive (including existence of additive inverses), such matrices are invertible. $\square$

Proof of Theorem 5.1. By Proposition 5.2, the functor (4.27) becomes the comparison from $[\mathcal{E}, \mathcal{X}]$ into the category of G -coalgebras. So the Theorem is now equivalent to comonadicity of the functor $\text{Lan}_J : [\mathcal{E}, \mathcal{X}] \longrightarrow [\mathcal{A}, \mathcal{X}]$. We already know that (4.27) is fully faithful. So it remains to prove that Lan_J preserves certain equalizers. In fact, it preserves all finite limits. Since limits in $[\mathcal{A}, \mathcal{X}]$ are formed pointwise, it suffices to see that each

$$\text{ev}_A \circ \text{Lan}_J : [\mathcal{E}, \mathcal{X}] \longrightarrow \mathcal{X}$$

preserves finite limits where $\text{ev}_A : [\mathcal{A}, \mathcal{X}] \longrightarrow \mathcal{X}$ is evaluation at $A \in \mathcal{A}$. By Lemma 4.1,

$$\text{ev}_A \circ \text{Lan}_J \cong \bigoplus_{U \leq A} \text{ev}_U .$$

Each evaluation ev_U preserves limits so their direct sum does. $\square$

6. SOME EXAMPLES

A *(set) species* in the sense of Joyal [9] is a functor $F : \mathfrak{S} \rightarrow \text{Set}$ where $\mathfrak{S}$ is the category of sets and bijective functions. A *pointed-set species* is a functor $F : \mathfrak{S} \rightarrow 1/\text{Set}$. An *R -module species* is a functor $F : \mathfrak{S} \rightarrow \text{Mod}^R$; the case where R is a field is the basic situation of [10].

Following [2], we write FI for the category of finite sets and injective functions. We then see that $\mathcal{M} = \text{FI}$ and $\mathcal{E} = \mathfrak{S}$, while $\mathcal{P} = \text{FI}_\#$, the category of injective partial functions.

Corollary 6.1. [2] *The functor*

$$\widehat{(-)} : [\mathfrak{S}, \text{Mod}^R] \rightarrow [\text{FI}_\#, \text{Mod}^R]$$

is an equivalence of categories.

Another example relevant to [13] is the category $\mathcal{A} = \text{FIVect}_{\mathbb{F}_q}$ of finite vector spaces over the field $\mathbb{F}_q$ of cardinality q (a prime power) and injective linear functions. Let $\mathcal{E} = \mathfrak{S}\mathfrak{I}_q$ be the category of finite $\mathbb{F}_q$ -vector spaces and linear isomorphisms. Then $\mathcal{P} = \text{FI}_\#^q$ is the category of finite $\mathbb{F}_q$ -vector spaces and injective partial linear functions.

Corollary 6.2. *The functor*

$$\widehat{(-)} : [\mathfrak{S}\mathfrak{I}_q, \text{Mod}^R] \rightarrow [\text{FI}_\#^q, \text{Mod}^R]$$

is an equivalence of categories.

Here are a few examples of categories $\mathcal{A}$ to which Theorem 5.1 applies with $\mathcal{E}$ the epimorphisms and $\mathcal{M}$ the monomorphisms:

- (a) the category of finite abelian groups and group morphisms;
- (b) the category of finite abelian p -groups and group morphisms;
- (c) the category of finite sets and all functions.

Theorem 5.1 also applies to $\mathcal{M}$ in place of $\mathcal{A}$ in these examples. Then $\mathcal{E}$ is replaced by the groupoid of invertible morphisms in $\mathcal{A}$. In case of example (a), the paper [8] describes the groupoid being represented in $\mathcal{X}$.

Consider the “algebraic” simplicial category Δ whose objects are all the natural numbers and whose morphisms $\xi : m \rightarrow n$ are order-preserving functions

$$\xi : \{0, 1, \dots, m-1\} \rightarrow \{0, 1, \dots, n-1\}.$$

Put $\mathcal{A} = \Delta^{\text{op}}$. Take $\mathcal{M}$ in $\mathcal{A}$ to consist of the surjections in Δ . Pushouts of surjections along arbitrary morphisms exist in Δ . Then $\mathcal{E} = \Delta_{\text{inj}}^{\text{op}}$ and $\mathcal{P}$ is the opposite of the category whose morphisms $m \rightarrow n$ are cospans

$$m \xrightarrow{\xi} r \xleftarrow{\sigma} n$$

in Δ with σ surjective. We could also take the “topological” simplicial category Δ_t (omit the object 0) to obtain a reinterpretation of the preoperads in $\mathcal{X}$ in the sense of [1].

Here is a rather trivial example involving Δ_t . Take $\mathcal{A}$ to be the category of non-empty ordinals and morphisms the order-preserving functions which preserve first element. Let $\mathcal{M}$ be the class of morphisms which are inclusions of initial segments.

This is part of a factorization system where $\mathcal{E} = \Delta_{\perp \neq \top}$ is the category of ordinals with distinct first and last element and morphisms the order-preserving functions which preserve first and last element. Sometimes $\mathcal{E}$ is called the category of *intervals*; there is a duality isomorphism

$$\mathcal{E} \cong \Delta_t^{\text{op}} .$$

In this case, not only do we have the equivalence

$$[\mathcal{E}, \mathcal{X}] \simeq [\mathcal{P}, \mathcal{X}]$$

of Theorem 5.1, we actually have an isomorphism

$$\mathcal{P} \cong \mathcal{E} .$$

Take $\mathcal{A}$ to be a (partially) ordered set with finite infima and the descending chain condition. Then every morphism is a monomorphism and the strong epimorphisms are equalities. So $\mathcal{E}$ is the discrete category $\text{ob}\mathcal{A}$ on the set of elements of the ordered set. The reader may like to investigate the case where $\mathcal{A}$ is the set of strictly positive integers ordered by division.

7. APPENDIX: REMARKS ON THE DOLD-PUPPE-KAN EQUIVALENCE

Let $\mathbb{D}$ be the additive category whose objects are the natural numbers, whose homs are

$$\mathbb{D}(m, n) = \begin{cases} \mathbb{Z} & \text{for } n = m, m+1 , \\ 0 & \text{otherwise} , \end{cases}$$

and whose composition

$$\mathbb{D}(n, p) \otimes \mathbb{D}(m, n) \longrightarrow \mathbb{D}(m, p)$$

is zero unless all three homs are $\mathbb{Z}$ in which case it is the canonical isomorphism $\mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}$. We write $\partial_n : n \longrightarrow n+1$ and $1_n : n \longrightarrow n$ for $1 \in \mathbb{Z}$.

For any additive category $\mathcal{V}$, the additive functor category $[\mathbb{D}^{\text{op}}, \mathcal{V}]$ is the category of differential non-negatively graded $\mathcal{V}$ -objects (or $\mathcal{V}$ -chain complexes).

Let Δ_{inj} denote the monoidal subcategory of the algebraic simplicial category Δ whose objects are all the natural numbers and whose morphisms $m \longrightarrow n$ are order-preserving injective functions

$$\mu : \{0, 1, \dots, m-1\} \longrightarrow \{0, 1, \dots, n-1\} .$$

This is the PRO for coaugmented objects: that is, objects equipped with a morphism from the tensor unit. Let $K : \Delta_{\text{inj}} \longrightarrow \Delta$ denote the inclusion.

There is a functor $J : \Delta_{\text{inj}} \longrightarrow \mathbb{D}$ which is the identity on objects and takes the injection $\partial_i : n-1 \longrightarrow n$ to 0 for $i < n$ and to ∂_n for $i = n$.

At the heart of the Dold-Puppe-Kan Theorem is the span of functors

$$\mathbb{D} \xleftarrow{J} \Delta_{\text{inj}} \xrightarrow{K} \Delta .$$

For any simplicial object $X : \Delta \longrightarrow \mathcal{V}$, consider the chain complex $N(X)$ defined as the right Kan extension of XK^{op} along J^{op} , which exists under a mild completeness

condition on $\mathcal{V}$.

$$\begin{array}{ccc}
 \Delta_{\text{inj}}^{\text{op}} & \xrightarrow{J^{\text{op}}} & \mathbb{D}^{\text{op}} \\
 K^{\text{op}} \downarrow & \xleftarrow{\rho} & \downarrow N(X) \\
 \Delta^{\text{op}} & \xrightarrow{X} & \mathcal{V}
 \end{array} \tag{7.28}$$

We have the weighted limit formula

$$N(X)_n = \lim(\mathbb{D}(J-, n), XK^{\text{op}}) .$$

The weight functor $\mathbb{D}(J-, n) : \Delta_{\text{inj}}^{\text{op}} \rightarrow \text{Ab}$ into the category Ab of abelian groups takes all objects to 0 except that $n-1$ and n go to $\mathbb{Z}$, and takes the morphisms $\partial_i : n-1 \rightarrow n$ to 0 for $0 \leq i < n$ and to the identity abelian group morphism $\mathbb{Z} \rightarrow \mathbb{Z}$ for $i = n$. The natural transformations $\mathbb{D}(J-, n) \Rightarrow SK^{\text{op}}$ for a simplicial abelian group S are clearly in bijection with elements $x \in S_n$ such that $d_i(x) = 0$ for $0 \leq i < n$. It follows that $N(X)_n$ is the intersection of the kernels of the $d_i : X_n \rightarrow X_{n-1}$ for $0 \leq i < n$. Also, $d_n : N(X)_n \rightarrow N(X)_{n-1}$ is induced by $d_n : X_n \rightarrow X_{n-1}$, making $N(X)$ a chain complex in $\mathcal{V}$.

We have the composite adjunction

$$\begin{array}{ccccc}
 [\mathbb{D}^{\text{op}}, \mathcal{V}] & \xrightarrow{\text{Res}_{J^{\text{op}}}} & [\Delta_{\text{inj}}^{\text{op}}, \mathcal{V}] & \xrightarrow{\text{Lan}_{K^{\text{op}}}} & [\Delta^{\text{op}}, \mathcal{V}] \\
 & \perp & & \perp & \\
 & \xleftarrow{\text{Ran}_{J^{\text{op}}}} & & \xleftarrow{\text{Res}_{K^{\text{op}}}} &
 \end{array}$$

where Res_H denotes restriction along H and its adjoints are left and right Kan extension along H . In our cases of H , the Kan extensions exist when $\mathcal{V}$ is finitely complete (and so has a bit of cocompleteness since it has finite direct sums).

Above we have identified the composite right-adjoint functor $\text{Ran}_{J^{\text{op}}}\text{Res}_{K^{\text{op}}}$ as the normalization functor N from simplicial objects to chain complexes. It is this functor that is proved an equivalence by Dold-Puppe-Kan [5, 6, 14].

Since K is strong monoidal, it follows from [3] that $\text{Lan}_{K^{\text{op}}}$ is strong monoidal for the convolution monoidal structures on $[\Delta^{\text{op}}, \mathcal{V}]$ and $[\Delta_{\text{inj}}^{\text{op}}, \mathcal{V}]$.

Although I have not completed the calculation, it seems that $\text{Res}_{J^{\text{op}}}$ is also strong monoidal for convolution monoidal structures on $[\Delta_{\text{inj}}^{\text{op}}, \mathcal{V}]$ and the usual tensor product on $[\mathbb{D}^{\text{op}}, \mathcal{V}]$. Since $\text{Res}_{J^{\text{op}}}$ is colimit preserving, we only need check the result on representables.

This would imply that $eN : [\Delta^{\text{op}}, \mathcal{V}] \rightarrow [\mathbb{D}^{\text{op}}, \mathcal{V}]$ is a monoidal equivalence.

The monoidal structure on Δ is not symmetric and so neither is the monoidal structure on $[\Delta^{\text{op}}, \text{Set}]$. As often happens, when we move to the additive context, a symmetry can be found: just transport the usual symmetry on chain complexes across the equivalence N .

There is something tantalizingly similar about the Dold-Puppe-Kan equivalence and the equivalence of Theorem 5.1. We mentioned towards the end of Section 6 that there is an isomorphism

$$\Delta_t^{\text{op}} \cong \Delta_{\perp \neq \top} .$$

From this viewpoint, the Dold-Puppe-Kan equivalence

$$N : [\Delta_{\perp \neq \top}, \mathcal{X}] \rightarrow [\mathbb{D}^{\text{op}}, \mathcal{X}]$$

is defined by

$$N(T)n = \bigcap_{0 \leq i < n} \ker(T(\sigma_i) : T(n+1) \longrightarrow Tn) .$$

The formula we have been working with for the right adjoint

$$\widetilde{(-)} : [\mathcal{P}, \mathcal{X}] \longrightarrow [\mathcal{E}, \mathcal{X}]$$

in the equivalence of Theorem 5.1 can be expressed as

$$\widetilde{TA} = \bigcap_{\text{non-invertible } m \in \mathcal{M}} \ker(T(m^*) : TA \longrightarrow TU) .$$

Surely these results are special cases of something more general.

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